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Zic2-associated holoprosencephaly is caused by a transient defect in the organizer region during gastrulation

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The putative transcription factor *ZIC2* is associated with a defect of forebrain development, known as Holoprosencephaly (HPE), in humans and mouse, yet the mechanism by which aberrant *ZIC2* function causes classical HPE is unexplained. The zinc finger domain of all mammalian *Zic* genes is highly homologous with that of the *Gli* genes, which are transcriptional mediators of Shh signalling. Mutations in *Shh* and many other Hh pathway members cause HPE and it has been proposed that *Zic2* acts within the Shh pathway to cause HPE. We have investigated the embryological cause of *Zic2*-associated HPE and the relationship between *Zic2* and the Shh pathway using mouse genetics. We show that *Zic2* does not interact with *Shh* to produce HPE. Moreover, molecular defects that are able to account for the HPE phenotype are present in *Zic2* mutants before the onset of Shh signalling. Mutation of *Zic2* causes HPE via a transient defect in the function of the organizer region at mid-gastrulation which causes an arrest in the development of the prechordal plate (PCP), a structure required for forebrain midline morphogenesis. The analysis provides genetic evidence that *Zic2* functions during organizer formation and that the PCP develops via a multi-step process.

INTRODUCTION

Holoprosencephaly (HPE) is the term used to describe a group of developmental disorders that feature a failure of forebrain midline development typified by the presence of fused cerebral hemispheres and incomplete separation of the eyes. It is the most common structural anomaly of the human forebrain. Although the majority of HPE cases are sporadic, autosomal dominant and autosomal recessive modes of inheritance have been described. In familial cases HPE shows incomplete penetrance and it is estimated that only 70% of obligate carriers show some clinical feature of HPE (1). Moreover, HPE shows variable expressivity with the same mutation being responsible for mild, moderate or severe HPE features in different individuals (2). Despite these non-Mendelian features of familial HPE, a total of nine genes have now been associated with HPE in humans (3–5). The genetics of HPE suggests that the establishment of the forebrain midline involves many genes whose level of activity must be tightly regulated and coordinated. Mouse mutations at these loci provide the opportunity to explore the embryological and molecular mechanisms of forebrain development and the interactions between HPE-associated genes.

Of the nine genes associated with HPE in humans the putative transcription factor *ZIC2* (MIM #609637) is unique in that it is associated with two distinct forms of HPE: the classical form and the middle interhemispheric variant (MIH) form (6,7). Recent work has provided an explanation of *Zic2* involvement in MIH HPE. In this HPE microform the anterior and posterior telencephalon forms relatively normally, but the mid-portion exhibits defective midline development and the choroid plexus fails to form. The roof-plate is a signalling centre that forms at the dorsal telencephalic midline and roof-plate ablation studies in the mouse show that BMP signals

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emanating from this region are required to prevent MIH HPE (8). A hypomorphic mutation of mouse *Zic2* that causes a mild form of HPE which most likely corresponds to MIH (9) exhibits a failure of roof-plate induction. The proposed model in which mild *Zic2* loss-of-function leads to disruption of roof-plate formation and ultimately to MIH HPE is supported by the fact that the only *ZIC2* mutation associated with MIH is a 12 amino acid in frame deletion C-terminal to the zinc finger domain (6). This mutation has been shown to generate a hypomorphic protein that elicits 60% of the normal ZIC2 transcriptional activation in in-vitro assays (10). While these studies clarify the role of *Zic2* with respect to MIH the more common association of *ZIC2* mutation with the classical form of HPE has remained unexplained.

The classical spectrum of HPE defects is likely to result from a defect in the generation of a functional Shh signal from a specialized structure in the ventral forebrain known as the prechordal plate (PCP). Human and animal studies have identified two developmental signalling pathways that are critical for the prevention of classical HPE: the Shh pathway and the TGF-β super-family pathway. Mutations that disrupt the formation or maintenance of the PCP cause HPE, although often in association with variable degrees of anterior truncation (11–16). The genes implicated in this aspect of ventral forebrain development mainly belong to the TGF-β super-family signalling pathway. Mutations in either *Shh* itself, or co-expressed molecules required for Shh function (*Cdon, Disp1*), allow the formation of the PCP, but result in a lack or depletion of Hh signal from the PCP (17–20). Genes, such as *LRP2* and *Smo*, which are associated with the response component of the Shh signalling pathway, when mutated, result in neurectoderm that is not competent to respond to the Shh signal and lead to HPE (21,22).

The *Zic* genes are vertebrate homologues of the drosophila pair rule gene *opa* (23). All *opa*-related genes have five Cys2, His2 zinc finger domains that are highly similar to *ci/Gli* genes and are assigned to the *Gli* super-family of transcription factors. The molecular mechanism of *Zic* gene function is poorly understood with few *in vivo* validated DNA-binding sequences or target genes and no proven association with developmental signalling pathways. Biochemical data, however, suggest a link with the Shh signalling pathway via interaction with the hedgehog transcriptional mediator *Gli*. All five mammalian Zic proteins (Zic1-5) physically interact via their zinc finger domain with Gli proteins and can bind to the *Gli* consensus DNA-binding site, either independently or synergistically with Gli (24). This has lead to the widely proposed hypothesis that in order to prevent classical HPE *Zic2* functions downstream of the Shh signal that emanates from the ventral forebrain (3,4,25).

To understand the basis of *Zic2* association with the classical form of HPE, we examined embryos homozygous for the kumba (*Ku*) allele of *Zic2* ($Zic2^{Ku}$ MGI:106679), which carries a missense mutation in the 4th zinc finger domain that abolishes the DNA binding and transcriptional activation ability of *Zic2* (10,26). These embryos exhibit a severe form of HPE without anterior truncation as do embryos null for *Shh*. Molecular markers show that ventral pattern and Shh signalling is compromised in the forebrain of $Zic2^{Ku/Ku}$ embryos. To investigate the relationship between Zic2 and the Shh pathway, we examined embryos carrying both the *Ku* allele of *Zic2* and a null allele of *Shh* ($Shh^{+/-}$; $Zic2^{Ku/+}$ or $Shh^{-/-}$; $Zic2^{Ku/Ku}$). These experiments found no evidence of a direct genetic interaction between *Zic2* and *Shh*, suggesting that *Zic2* does not act downstream of *Shh* to cause HPE. Instead we find that at 9.5 days post-coitum (dpc) in *Zic2* mutant embryos the PCP and anterior notochord are compromised and that this is presaged by overt disruption of these tissues at the early headfold stage of development. Investigations of gastrulation stage embryos show that the initial stages of anterior mesendoderm allocation (including formation of the PCP) occur relatively normally in $Zic2^{Ku/Ku}$ embryos, as does trunk notochord production. At the late streak stage, however, the organizer region (or node) is abnormal and insufficient anterior notochord precursors are generated. The dearth of anterior notochord cells apparently leads to an absence of molecularly normal PCP cells by the headfold stage of development. *Zic2* is therefore required before the onset of Shh signalling in the mouse embryo and *Zic2* mutation causes HPE via a previously unsuspected mechanism. The $Zic2^{Ku}$ allele not only explains *Zic2* involvement in HPE, but provides new insight into organizer function and PCP development.

RESULTS

zic2 mutation causes a severe form of HPE

Embryos homozygous for the *Ku* allele of *Zic2* have a variety of defects including delayed neural tube closure and delayed neural crest production (26). In addition, at 9.5 dpc *Ku* mutant embryos have a visibly abnormal forebrain, perhaps indicative that this *Zic2* mutation may cause HPE. To determine whether $Zic2^{Ku/Ku}$ mutants display HPE features, embryos were examined at 12.5 dpc. At this stage $Zic2^{Ku/Ku}$ embryos have grossly abnormal cranial development. Most homozygous embryos have exencephaly and all homozygotes show abnormally spaced eyes with dysmorphologies ranging from hypotelorism (37%, $N = 19$) to cyclopia (58%, $N = 19$), with one embryo exhibiting two eyes fused at the midline (5%, $N = 19$) (Fig. 1). The remnants of the forebrain tissue are found rostral of the eye and this tissue remains as a single mass of neural cells with no evidence of the interhemispheric fissure that would normally separate the left and right hemispheres (Fig. 1C–F). The *Ku* allele of *Zic2* therefore causes the typical spectrum of HPE phenotypes and in this respect resembles the forebrain phenotype of $Shh^{-/-}$ embryos (17).

shh and *zic2* do not interact genetically to cause HPE

As described in the Introduction, the sequence and functional homologies of the *Gli* and *Zic* genes have led to the hypothesis that *Zic2* is involved in mediating the *Shh* signal within the mouse forebrain. To examine the relationship between *Zic2* and *Shh*, we carried out a series of genetic experiments. Initially, embryos heterozygous for mutations at both *Zic2* and *Shh* ($Shh^{+/-}$; $Zic2^{Ku/+}$) were generated. A cross between $Shh^{+/-}$ and $Zic2^{Ku/+}$ animals yielded viable *trans*-heterozygote progeny in the expected proportion (27%; $N = 296$; $P > 0.05$ using the χ^2 test). To confirm that *trans*-heterozygotes

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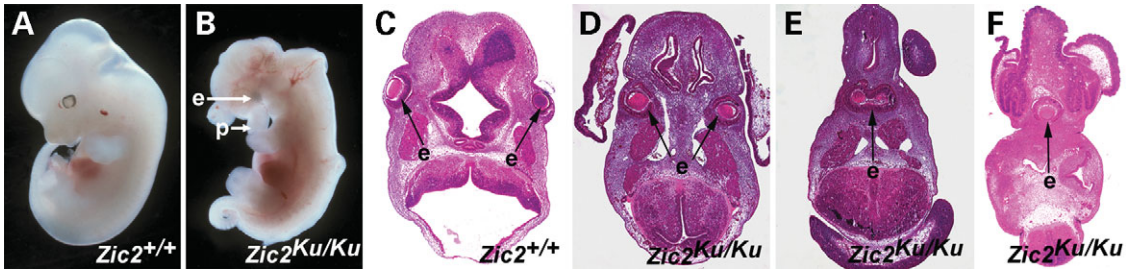


Figure 1. *Zic2* mutation causes cyclopia with proboscis. Lateral view of intact embryos or transverse sections through the head of 12.5 dpc embryos. (**A**) A wild-type embryo. (**B**) A $Zic2^{Ku/Ku}$ embryo with exencephaly, spina bifida, a looped tail, an internally located eye and a proboscis. (**C**) A wild-type embryo with two, externally located eyes and clearly divided L and R hemispheres. (**D**) A $Zic2^{Ku/Ku}$ embryo with internal and closely spaced eyes (hypotelorism). (**E**) A $Zic2^{Ku/Ku}$ embryo with internal and incompletely separated eyes (synophthalmia). (**F**) A $Zic2^{Ku/Ku}$ embryo with one centrally located eye (cyclopia). e, eye; p, proboscis.

did not exhibit the defects characteristic of either $Shh^{-/-}$ or $Zic2^{Ku/Ku}$ mutants, 9.5 dpc embryos were recovered from a further set of crosses and the *trans*-heterozygotes examined via whole mount *in situ* hybridization (WMISH) with a variety of markers. The markers included *Shh*, *T*, *Zic2*, *Fgf8, Foxa2* and *Foxg1*, and no differences were observed between *trans*-heterozygous embryos and stage-matched, wild-type littermates (data not shown).

To explore the consequence of other compound genotypes *trans*-heterozygous animals were inter-crossed and a total of 309 embryos at stages between 7.5 and 10.5 dpc were recovered and genotyped. All nine genotype classes were present at the expected ratio (data not shown). Embryos were examined visually and using WMISH to the forebrain markers *Fgf8* and *Foxg1*. In all cases, embryos of the genotype $Shh^{+/-}$; $Zic2^{Ku/Ku}$ fell into the phenotypic range of $Shh^{+/+}$; $Zic2^{Ku/Ku}$ embryos. Similarly, $Shh^{-/-}$; $Zic2^{Ku/+}$ embryos resembled those of the genotype $Shh^{-/-}$; $Zic2^{+/+}$ (Fig. 2). In contrast, embryos that were homozygous for both mutations exhibited a phenotype distinct from that of either $Shh^{-/-}$ or $Zic2^{Ku/Ku}$ embryos. At 9.5 dpc the forebrain of double mutant embryos ($Shh^{-/-}$; $Zic2^{Ku/Ku}$) is severely truncated as demonstrated by the depletion or loss of the *Fgf8* and *Foxg1* expression domains (Fig. 2D and J). By 10.5 dpc the forebrain of these embryos has formed a fluid-filled, cyst-like structure. It can also be seen that they are small, have failed to turn and have an obvious heart defect (Fig. 2M). The finding that mutation of either gene is not sensitive to a decreased dose of the other, in combination with the novel phenotype of the double mutants, is not consistent with the proposed model in which Zic2 acts to mediate the Shh signal in the 9.5 dpc brain. The remaining experiments were therefore directed at understanding how HPE occurs in $Zic2^{Ku/Ku}$ embryos.

Shh signalling in the forebrain is abnormal in *zic2*$^{ku/ku}$ embryos

Severe forms of HPE are generally associated with a failure of *Shh* signalling in the ventral forebrain and analysis of the mutants revealed that $Zic2^{Ku/Ku}$ embryos exhibited a midline defect as shown by the reduced expression domain of *Fgf8* in the anterior neural ridge (Fig. 3A–D). This analysis, along with that of the double mutants shown in Fig. 2, highlighted the morphological and molecular variability of mutant embryos at 9.5 dpc. For example, the embryos shown in Figs. 2L and 3B have a reduced domain of *Fgf8* expression at the anterior midline, whereas the embryos in Figs. 2I and 3D lack expression of *Fgf8* in this region. This phenotypic variability presumably presages the variable eye spacing described in 12.5 dpc mutants above. To further examine the forebrain midline, embryos were sectioned after WMISH to *Foxg1*. This confirmed that the ventral forebrain tissue between the optic eminences is greatly reduced in $Zic2^{Ku/Ku}$ embryos (Fig. 3E and F). To determine if *Shh* signalling is intact in the forebrain of mutant embryos, we examined the expression pattern of three genes known to be transcriptionally responsive to Shh signalling. Both the homeodomain containing transcription factor *Titf1* (formerly *Nkx2.1*) and the zinc finger containing transcription factor *Gli1* are expressed in the ventral telencephalon in a Shh-dependent manner (27–29). In $Zic2^{Ku/Ku}$ embryos the expression domain of both of these genes remains unaffected (Fig. 3G–J). In contrast, the expression of the Shh receptor *Ptch1* which is positively regulated by Shh signalling (30) is lost from the telencephalon of mutant embryos (Fig. 3K and L) raising the possibility that Shh signalling is compromised in the forebrain of $Zic2^{Ku/Ku}$ embryos.

To clarify whether the observed effects on Shh-dependent gene expression reveal a role for the dorsally expressed *Zic2* gene in the Shh-signalling pathway, we examined *Shh* expression. If, as predicted, the dorsally expressed *Zic2* acts downstream of *Shh*, then *Shh* expression in the 9.5 dpc forebrain of $Zic2^{Ku/Ku}$ embryos should be intact. We find, however, that the expression of *Shh* is also altered in mutant embryos in a manner reminiscent of *Ptch1* expression. As shown in Figure 3M and N *Shh* transcripts are absent from the ventral telencephalon of $Zic2^{Ku/Ku}$ embryos. The wild-type telencephalon expression of *Shh* marks a structure known as the PCP. These expression data concur with the genetic interaction studies and suggest that *Zic2* acts to establish *Shh* expression in the PCP.

zic2 mutation causes aberrant PCP and anterior notochord development

One possible explanation of the lack of PCP *Shh* expression is that *Zic2* is required to regulate the transcription of *Shh* in the PCP. This seems unlikely, however, since *Zic2* and Shh are not coexpressed within the PCP (31,32). It is also possible that the depleted PCP expression of *Shh* indicates a more general

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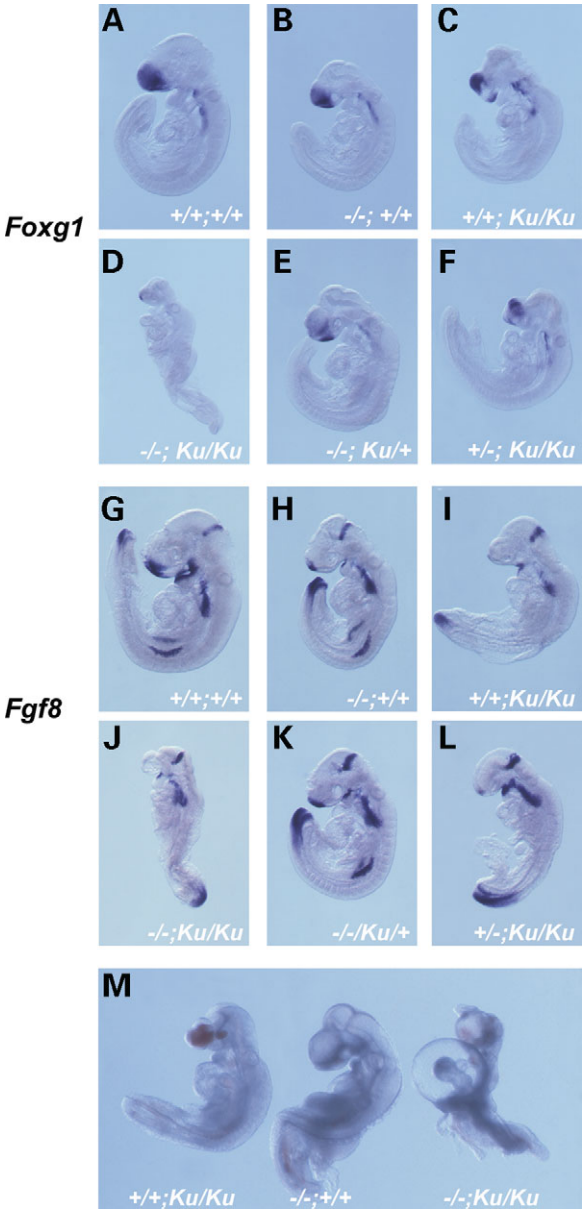


Figure 2. *Shh* and *Zic2* do not interact in the murine forebrain. Lateral view of compound mutants, anterior is to the left. (**A**–**F**) The 9.5 dpc embryos after WMISH to *Foxg1*. Decreasing the dose of *Zic2* does not alter the $Shh^{-/-}$ phenotype; compare (B) and (E). Similarly, decreasing the dose of *Shh* on the *Zic2* mutant background does not alter the phenotype; compare (C) and (F). The double mutant, in (D), has a greatly diminished forebrain exhibiting little *Foxg1* expression. (**G**–**L**) The 9.5 dpc embryos after WMISH to *Fgf8*. Decreasing the dose of *Zic2* does not alter the $Shh^{-/-}$ phenotype; compare (H) and (K). Similarly, decreasing the dose of *Shh* on the *Zic2* mutant background does not alter the phenotype; compare (I) and (L). The double mutant, in (J), has a greatly diminished forebrain exhibiting no *Fgf8* expression. (**M**) The 10.5 dpc embryos, the double mutant embryo on the right has a fluid filled cyst-like structure in place of the forebrain, it also has an enlarged heart cavity, is small, posteriorly truncated and has failed to turn.

failure of midline development and this is supported by the evident disruption at 9.5 dpc of another midline structure, the anterior notochord. The failure of notochord development can be seen in the hindbrain region of the embryos hybridized to *Shh*. At this axial level *Shh* is expressed in both the ventral region of the neural tube (the floorplate) and in the underlying notochord. The WMISH analysis shows that in mutant embryos the floorplate expression of *Shh* remains relatively intact but the notochord appears to consist only of dispersed clumps of *Shh* expressing cells (Fig. 3M–P). This is shown more clearly after WMISH to the T-box transcription factor *T* which is expressed only in the notochord. Sectioning of these embryos shows that the regions of the embryo that fail to hybridize to the *T* probe in the WMISH hybridization experiment lack a notochord (Fig. 3Q–T).

At 9.5 dpc there are few markers of the PCP, but at 8.0 dpc, during the headfold stages of development, the PCP can be defined by the co-expression of *Shh*, *Foxa2* and *Gsc* (33,34). In $Zic2^{Ku/Ku}$ embryos, at the headfold stage, the expression of each of these genes is depleted or absent in the PCP region, relative to stage matched wild-type controls (Fig. 4A–F). This demonstrates that *Zic2* acts at an early stage of PCP development and therefore acts earlier during forebrain development than previously recognized. Additionally, at the headfold stage two of these genes (*Shh* and *Foxa2*) are also expressed in the emerging notochord. As shown in Fig. 4, the expression of these two genes, and of the notochord specific marker *T*, is downregulated or discontinuous, indicating that the anterior notochord is also affected at this early stage of development.

PCP development is initiated in $zic2^{ku/ku}$ mutants

To identify the primary role of *Zic2* during development of the anterior embryonic midline, we investigated *Zic2* mutants at earlier stages. *Zic2* is expressed prior to and during gastrulation (32) and it is possible that it plays a role in the earliest steps of anterior axis formation. The orientation of the anterior–posterior embryonic axis is dependent upon the location of the anterior visceral endoderm (AVE) prior to gastrulation (35). Examination of *Hex* (Fig. 5A) and *Hesx1* (not shown) expression in pre-gastrulation $Zic2^{Ku/Ku}$ mutants demonstrates that the visceral endoderm undergoes appropriate anterior movement with the AVE located at the embryonic/extraembryonic junction. Additionally, prior to gastrulation *T* expression is present in a ring at the embryonic/extraembryonic junction. In mutant embryos *T* transcripts become restricted to the posterior of the embryo and are found as expected in the elongating primitive streak (Fig. 5B and C). In embryos that fail to complete these early stages of A–P axis formation precocious differentiation of the epiblast occurs (36), whereas in $Zic2^{Ku/Ku}$ embryos the epiblast follows the normal pattern of differentiation as determined by the correct expression of the pluripotent cell marker *Oct4* (data not shown). This analysis shows that the earliest stages of A–P axis formation occur correctly in *Zic2* mutants.

As gastrulation proceeds the cells at the anterior of the primitive streak form an organizer region that will generate the anterior mesendoderm cells. The cells of this early gastrula organizer give rise to the PCP precursors and at ES–MS stages they can be identified via *Foxa2* expression. In ES and MS embryos *Foxa2* expression is equivalent in stage-matched wild-type and *Zic2* mutant embryos (Fig. 5D). Additionally, WMISH to *Lhx1*, *Cer1*, *Hhex* (Fig. 5E–G) and

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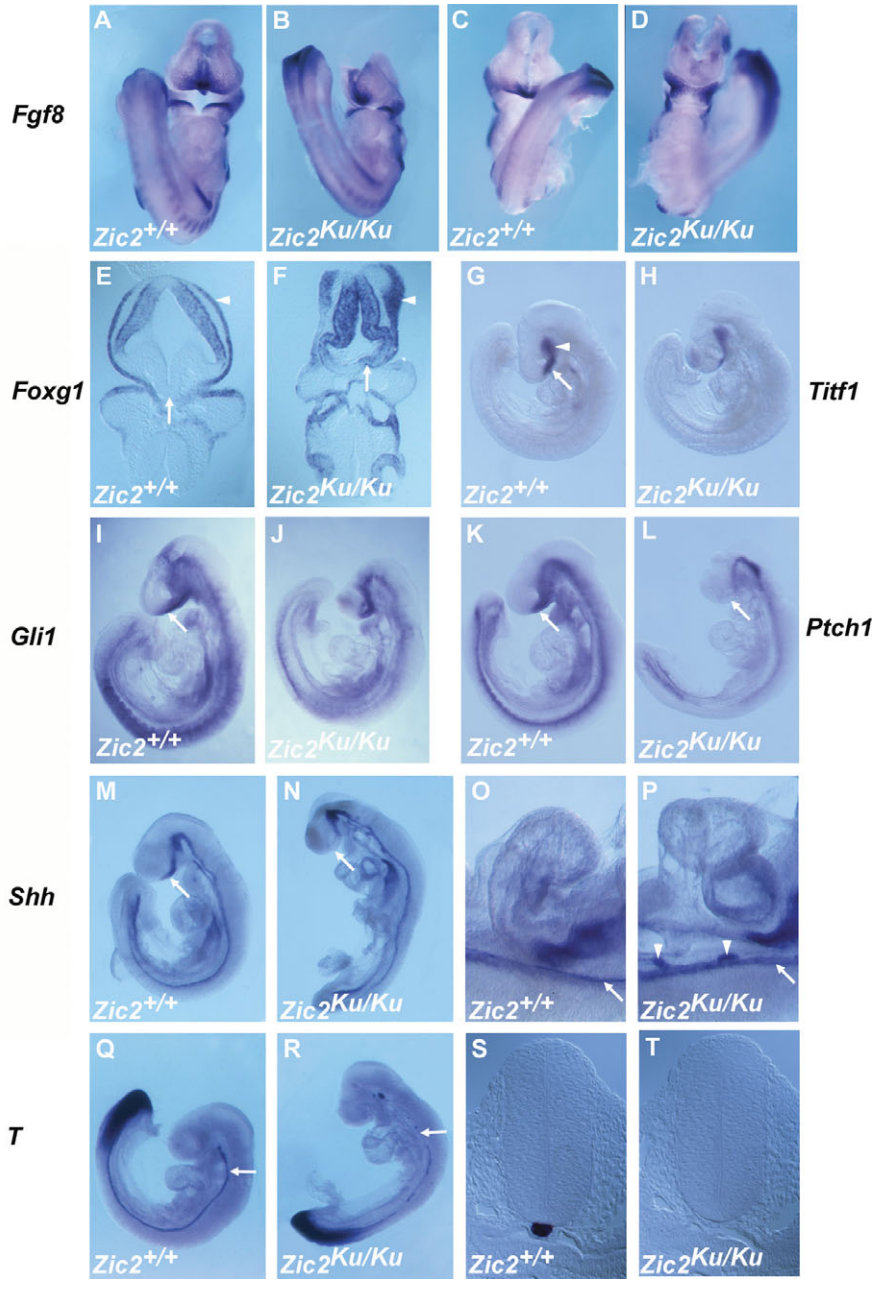


Figure 3. Shh signalling is down-regulated in the forebrain. The embryos in (**A**–**D**) are viewed from the anterior, whereas all other intact embryos are shown in lateral view. All sections are transverse. (A and B) Stage-matched 9.5 dpc embryos after WMISH to *Fgf8*, the *Zic2*$^{Ku/Ku}$ embryo retains some *Fgf8* expression in the anterior neural ridge. (C and D) A second pair of stage-matched 9.5 dpc embryos after WMISH to *Fgf8*, the *Zic2*$^{Ku/Ku}$ embryo exhibits no *Fgf8* expression in the anterior neural ridge. (**E**) A 16 somite wild-type embryo after WMISH to *Foxg1*. (**F**) A 16 somite *Zic2*$^{Ku/Ku}$ embryo after WMISH to *Foxg1*. The surface ectoderm is thickened (arrowhead) and the tissue between the optic eminences is lost (arrow). (**G**) A 19 somite wild-type embryo showing the telencephalic (arrow) and diencephalic (arrow head) expression domains of *Titf1*. (**H**) A 19 somite *Zic2*$^{Ku/Ku}$ embryo showing unaltered *Titf1* expression. (**I**) A 19 somite wild-type embryo showing the telencephalic (arrow) expression domain of *Gli1*. (**J**) A 19 somite *Zic2*$^{Ku/Ku}$ embryo showing unaltered *Gli1* expression in the telencephalon. (**K**) A 15 somite wild-type embryo showing the telencephalic (arrow) expression domain of *Ptch1*. (**L**) A 15 somite *Zic2*$^{Ku/Ku}$ embryo. *Ptch1* signal is absent from the telencephalon (arrow). (**M**) A 20 somite wild-type embryo showing the telencephalic (arrow) expression domain of *Shh*. (**N**) A 20 somite *Zic2*$^{Ku/Ku}$ embryo in which telencephalic *Shh* signal is absent (arrow). (**O**) Enlarged view of the wild-type embryo shown in (M) at the level of the heart. *Shh* expression marks both the floor plate of the neural tube and the overlying notochord (arrow). (**P**) Enlarged view of the *Zic2*$^{Ku/Ku}$ embryo shown in (N) at the level of the heart. *Shh* expression is maintained in the floor plate of the neural tube (arrow), whereas the notochord consists of discontinuous clumps of *Shh* expressing cells (arrowheads). (**Q**) A 17 somite wild-type embryo showing *T* expression in the notochord (arrow). (**R**) A 16 somite *Zic2*$^{Ku/Ku}$ embryo. *T* signal is discontinuous in the notochord (arrow). (**S**) A section of the wild-type embryo in (Q). The notochord is clearly visible. (**T**) A section of the *Zic2*$^{Ku/Ku}$ embryo in (R). The notochord is absent.

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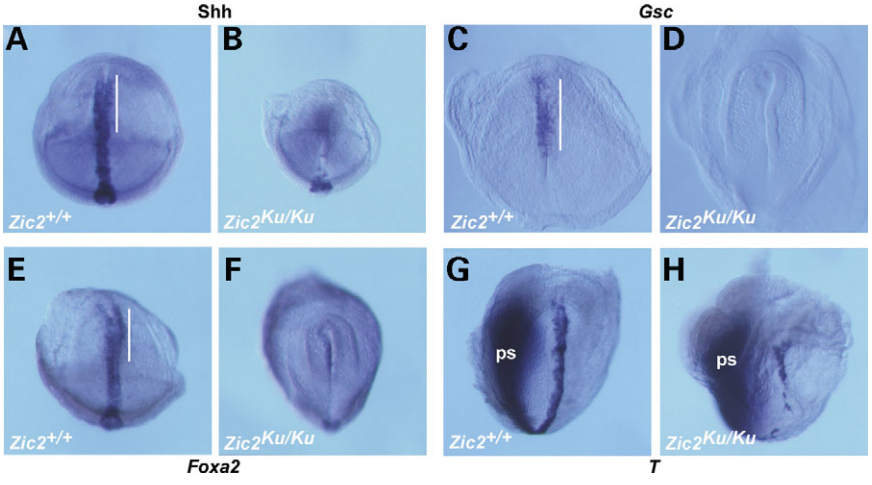


Figure 4. Anterior mesendoderm is depleted at the headfold stage of development. Anterior view of embryos at the headfold stage of development after WMISH to the genes shown to the top or bottom of the respective panels. The white line in (**A**), (**C**) and (**E**) corresponds to the extent of the PCP. (A) An early-headfold stage wild-type embryo showing that *Shh* expression extends throughout the PCP and the anterior notochord. (**B**) An early-headfold stage $Zic2^{Ku/Ku}$ embryo in which *Shh* expression is absent from the PCP and from much of the anterior notochord. (C) An early-headfold stage wild-type embryo showing that *Gsc* expression is restricted to the PCP. (**D**) An early-headfold stage $Zic2^{Ku/Ku}$ embryo in which *Gsc* expression is absent from the PCP. (E) An early-headfold stage wild-type embryo showing that *Foxa2* expression extends throughout the PCP and the anterior notochord. (**F**) An early-headfold stage $Zic2^{Ku/Ku}$ embryo in which *Foxa2* expression is absent from the PCP and from much of the anterior notochord. (**G**) A late-headfold stage wild-type embryo showing *T* expression in the primitive streak and notochord of the cranial region. (**H**) A late-headfold stage $Zic2^{Ku/Ku}$ embryo in which *T* expression in the cranial notochord is discontinuous. ps, primitive streak.

Gsc (data not shown) revealed that the expression of genes in the emerging anterior mesendoderm, the definitive endoderm and lateral mesoderm are comparable between mutant and wild-type embryos at the LS stage. At this stage of development, however, gene expression at the anterior streak begins to differ in the mutant embryos (Fig. 5G). This feature becomes more apparent at a slightly later stage of gastrulation (OB) once the primitive streak reaches the bottom of the egg cylinder and the node becomes visible. The node is now the site of organizer activity and it is at this stage that the first consistent signs of aberrant gene expression in *Zic2* mutants are found (Fig. 5H–J). At this mid-gastrula stage markers of the node, the endoderm emerging from the node and the mesoderm surrounding the node show subtle differences in expression. In contrast, gene expression at the anterior of the embryo (in the PCP precursors) is maintained. This is best demonstrated by the expression of *Hhex*, which is markedly down-regulated at the node, but maintained in both the blood islands of the extraembryonic mesoderm and the anterior mesendoderm (Fig. 5J). These data imply that the PCP cells are specified correctly, but that the node of the mid-gastrula embryo is defective.

Since PCP cells are detected via molecular markers at this mid-gastrula stage, yet not at the early headfold stage, we examined intervening stages to chart the apparent breakdown of the PCP. At both the EB and LB stages we find that some PCP markers are expressed while others are not. For example, at the EB stage expression in the anterior mesendoderm of *Hhex* is maintained, whereas *Foxa2* transcripts are lacking (Fig. 5O and M). In EB embryos *Hesx1* and *Hhex* serve as markers of mesendoderm and these are both still expressed (Fig. 5R and data not shown), whereas *Shh* signal is clearly lacking (Fig. 5Q). Evidently, PCP cells are specified, but their identity is not maintained and they fail to express PCP characteristic markers by the early headfold stage (as shown by the lack of *Shh*, *Foxa2* and *Gsc* expression in Fig. 4).

The anterior notochord is depleted in *zic2$^{ku/ku}$* mutants

The WMISH analysis of mid-gastrula embryos established that at this stage the node itself is abnormal and a decreased number of differentiated cells emerge from the node. The cells that pass through the mid-gastrula node will give rise to the anterior notochord. At 7.5 dpc (EB stage) the cells of the anterior notochord are marked by several genes including *T*, *Chrd*, *Foxa2*, *Shh*, *Hhex*. The expression of each of these is specifically down-regulated in the emerging notochord of EB stage *Zic2* mutants (Fig. 5K–O) implying that the future discontinuities arise due to a lack of specification. In contrast, development of the neurectoderm proceeds relatively normally, with *Otx2* expression becoming restricted to the neurectoderm as it differentiates *in situ* (Fig. 5P).

The mesendoderm defects occur prior to the requirement for Hh signalling

It remains possible that *Zic2*-associated HPE arises due to an early interaction between *Shh* and *Zic2*. Our characterization of the *Zic2* mutant phenotype implies that this is unlikely since the earliest molecular abnormality detected during the analysis of the $Zic2^{Ku/Ku}$ mutants occurred at ~7.0 dpc in the node and emerging mesendoderm of the LS and OB embryo (Figs. 5G–J and 6A). In contrast, the earliest stage at which *Shh* expression is detected in the mouse embryo using WMISH is some 12 h later, during the EB stage of development at ~7.5 dpc (see the wild-type embryo in Fig. 5N). It is, however, possible that at LS/OB stages *Shh* transcripts are present at a level below WMISH detection or that other

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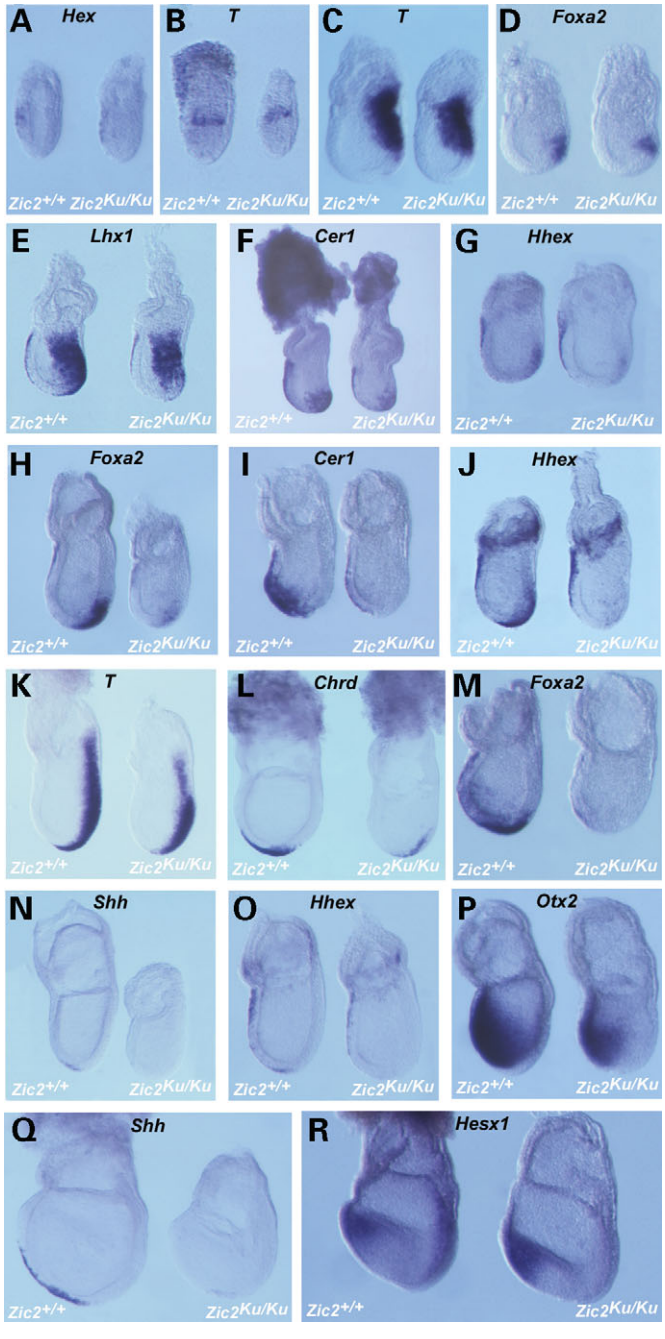


Figure 5. The mid-gastrula node is defective in *Zic2* mutants. Lateral views of embryos after WMISH; anterior is to the left, except in (**B**) where the orientation of the embryo cannot be determined. (**A**) H*hex* expression shows that the AVE is correctly positioned in PS $Zic2^{Ku/Ku}$ embryos. (B) *T* expression in PS stage embryos shows that the proximo-distal axis is correctly aligned in $Zic2^{Ku/Ku}$ embryos. (**C**) *T* expression in MS stage embryos shows that initial mesoderm formation occurs in the *Zic2* mutant. (**D**) *Foxa2* expression in MS stage embryos; the organizer region forms in $Zic2^{Ku/Ku}$ embryos. (**E**) *Lhx1* expression is equivalent in the lateral mesoderm and anterior endoderm of LS stage embryos. (**F**) *Cer1* expression is unaltered in the definitive endoderm of LS stage $Zic2^{Ku/Ku}$ embryos. (**G**) Hhex signal is depleted in the node but not PCP of LS stage $Zic2^{Ku/Ku}$ embryos. (**H**) *Foxa2* signal is depleted at the node of OB stage *Zic2* mutants. (**I**) *Cer1* signal is reduced in the endoderm emerging from the node, but not the PCP of OB stage *Zic2* mutant embryos. (**J**) *Hhex* signal is diminished in the node, but not the PCP or blood islands of the OB stage *Zic2* mutant. (**K**) *T* expression is decreased in the anterior notochord and posterior primitive streak of the EB stage *Zic2* mutant. (**L**) *Chrd* signal is absent from the anterior notochord and decreased in the node of the EB stage $Zic2^{Ku/Ku}$ embryos. (**M**) *Foxa2* transcription is down-regulated in the PCP and node of the EB stage *Zic2* mutant. (**N**) *Shh* expression is absent from the anterior notochord of the EB stage *Zic2* mutant. (**O**) *Hhex* expression is maintained in the PCP but lost from the anterior notochord in the EB stage $Zic2^{Ku/Ku}$ embryos. (**P**) *Otx2* signal is unaltered in the neurectoderm of the EB stage *Zic2* mutant. (**Q**) *Shh* expression is largely depleted in the anterior mesendoderm of the LB stage *Zic2* mutant. (**R**) *Hesx1* expression is maintained in the neurectoderm of the LB stage *Zic2* mutant embryo.

Hh-related molecules are expressed at this early stage of development. To determine whether defects in Hh signalling also affect node development, we examined the 7.0 dpc node of embryos null for either *Shh* itself or for the Hh receptor smoothened (*Smo*). Since *Smo* is the receptor for all mammalian hedgehogs no Hh signalling takes place in $Smo^{-/-}$ embryos (37). In either case, the expression of *Foxa2* at the node is normal in the mutant embryos relative to wild-type stage-matched littermates, unlike in *Zic2* mutants that show a noticeable decrease in *Foxa2* signal (Fig. 6). This experiment demonstrates that the origin of the *Zic2*-associated HPE phenotype is independent of Hh signalling.

The mesendoderm defects are not caused by aberrant cell death or proliferation

Potentially, lack of proliferation, increased cell death or impaired differentiation may be the cause of insufficient anterior notochord cells. We therefore examined proliferation and cell death in wild-type and stage-matched *Zic2* mutant embryos at two different time-points (Table 1 and Fig. 7). These were OB stage (the stage of onset of the molecular phenotype detected in our WMISH analysis) and EB stage (12 h after phenotype onset). The number of proliferating and apoptotic cells was counted in the embryonic portion of embryos. Counts were obtained for each of the three germ layers (ecto-, meso- and endoderm) and were also pooled to generate a total count for each embryo analysed. At the earlier OB stage no significant differences were observed between wild-type and $Zic2^{Ku/Ku}$ embryos. At the EB stage, however, there was a significant increase in the number of apoptotic cells in the ectoderm of $Zic2^{Ku/Ku}$ embryos (Table 1). This difference was distributed throughout the ectoderm and because it occurs after the onset of the molecular phenotype we conclude that it is a consequence, rather than a cause, of the phenotype.

DISCUSSION

The primary defect in *zic2*-associated classical HPE is independent of Shh signalling

The expression of *Zic2* within the neurectoderm and the high homology between *Zic* and *Gli* genes have led to the hypothesis that *Zic2* acts downstream of Shh signalling in the forebrain (3,4,25). In this study, the first to directly test this hypothesis, we have performed an in depth analysis of mouse embryos homozygous for a strong allele of *Zic2* which results in the most severe form of HPE known as cyclopia. In these embryos we find that Shh signalling is indeed disrupted in the forebrain, but this is due to a defect of the tissue which is the normal source of the Shh signal (the PCP). HPE is therefore sec-

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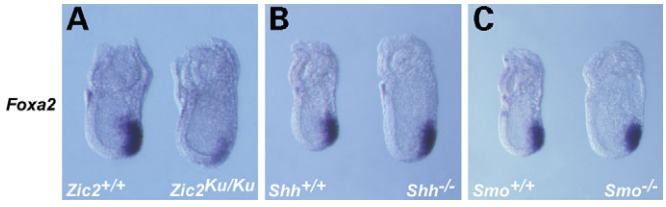


Figure 6. Loss of Hedgehog signalling does not affect the mid-gastrula node. Lateral views of LS-stage embryos after WMISH to the node expressed *Foxa2* gene, anterior is to the left. (**A**) A wild-type embryo and a $Zic2^{Ku/Ku}$ embryo showing that the node expression of *Foxa2* is depleted in the mutant embryo. (**B**) A wild-type embryo and a $Shh^{-/-}$ embryo showing that the node expression of *Foxa2* is unaltered in the mutant embryo. (**C**) A wild-type embryo and a $Smo^{-/-}$ embryo showing that the node expression of *Foxa2* is maintained in the absence of Hedgehog signalling.

Table 1. The number and distribution of mitotically active or apoptotic cells in gastrulation stage embryos

Embryo	Ectoderm	Mesoderm	Endoderm	Total
Proliferation (histone H3 positive)				
OB $Zic2^{+/+}$	217.6 ± 137.1	60.8 ± 42.5	43.1 ± 34.2	321.4 ± 205.3
OB $Zic2^{Ku/Ku}$	242.0 ± 50.6	81.3 ± 45.4	47.9 ± 13.1	371.3 ± 80.0
EB $Zic2^{+/+}$	438.6 ± 164.3	240.6 ± 62.4	95.0 ± 23.5	774.2 ± 226.9
EB $Zic2^{Ku/Ku}$	308.8 ± 92.8	183.6 ± 61.8	75.6 ± 42.8	568.0 ± 150.5
Apoptosis (Caspase 3 positive)				
OB $Zic2^{+/+}$	23.0 ± 14.7	4.0 ± 7.2	1.2 ± 1.3	28.2 ± 28.2
OB $Zic2^{Ku/Ku}$	33.9 ± 25.1	13.8 ± 18.9	3.0 ± 3.6	50.6 ± 32.9
EB $Zic2^{+/+}$	17.4 ± 11.08	9.8 ± 9.6	0.6 ± 0.9	27.8 ± 18.6
EB $Zic2^{Ku/Ku}$	86.2 ± 49.3[a]	17.2 ± 19.8	8.0 ± 7.7	111.4 ± 74.1

Embryos were double-hybridized with antibodies enabling cell counts for both mitosis and apoptosis on the same embryo (OB $Zic2^{+/+}$: $N = 10$, OB $Zic2^{Ku/Ku}$: $N = 8$, EB $Zic2^{+/+}$: $N = 5$, EB $Zic2^{Ku/Ku}$: $N = 5$). Counts were pooled within genotype classes and the mean number of cells per germ layer and in the total embryonic region calculated. Data are shown as the mean ± SEM values.
[a]Number of cells is significantly different to the corresponding wild-type data set, $P < 0.05$ (two-tailed t test).

ondary to the primary function of *Zic2* which is to ensure the generation of a functional PCP. Additionally, we have used compound mutants to show that embryos homozygous for mutations at both *Shh* and *Zic2* do not mimic the $Shh^{-/-}$ forebrain phenotype, again indicating that mutation of *Zic2* does not cause HPE because of a requirement for Zic2 downstream of Shh signalling in the murine forebrain. The defect that underlies *Zic2*-associated HPE can be traced to mid-gastrulation at which stage the expression of several genes in the organizer region of the embryo is down-regulated. Embryos that are either null for *Shh* or for the Hh receptor *Smo* (and therefore lack all Hh signalling) do not have a defective organizer region at mid-gastrulation, ruling out the possibility that Shh signalling is required for the primary function of *Zic2* revealed by our experiments. Our experiments do not preclude interaction between *Zic2* and *Shh* at some other stage or developmental process, for example it is possible that the extreme phenotype of the compound homozygous embryos is due to interactions between these proteins outside of the PCP.

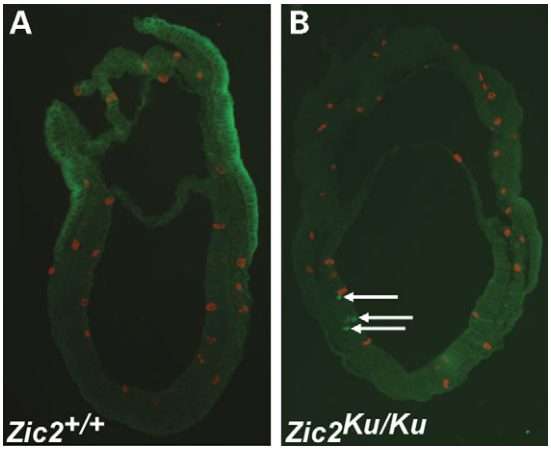


Figure 7. Increased apoptosis in EB stage mutant embryos. Representative longitudinal sections through EB stage embryos. (**A**) A section of a wild-type embryo. (**B**) A section of a $Zic2^{Ku/Ku}$ embryo. Histone H3 positive cells are red and Cleaved Caspase three positive cells are green (indicated by arrows). The images are at identical scale.

zic2-associated classical HPE is caused by a failure of PCP development

At the headfold stage of development the PCP can be defined molecularly by the co-expression of *Gsc*, *Shh* and *Foxa2* (33,34). Investigation of $Zic2^{Ku/Ku}$ embryos demonstrates that there are no cells at the anterior midline of headfold stage embryos which express this combination of markers, indicating that the PCP cells have either degenerated or undergone a fate transition, and it is well documented that failure to develop a functioning PCP will cause HPE. The origin of PCP failure in $Zic2^{Ku/Ku}$ embryos can be traced to the gastrulation stage of development. In mutant embryos the initial establishment of the anterior axis proceeds as normal. The AVE is correctly positioned prior to gastrulation and after the onset of gastrulation the anterior definitive endoderm (ADE) that includes the PCP precursor cells is produced relatively normally as shown by the expression of *Lhx1*, *Cer1, Hhex* and *Foxa2* (Fig. 5E–H). Although the *in situ* analysis does not rule out a small deficit in the number of PCP cells produced, the ADE and PCP cells arise from the organizer of the early-gastrula (ES and MS stages) (38), which is marked by the expression of *Foxa2*, and this appears normal in the mutant embryos supporting the interpretation that PCP formation is initiated in $Zic2^{Ku/Ku}$ embryos. At mid-gastrula stages (LS and OB stages), the cells progressing through the organizer region will contribute predominantly to the cranial notochord (38). In mutant embryos this transition is either delayed or does not occur such that the production of the mesendoderm is disrupted for a period of time. The first identified molecular and functional abnormalities in the mutant embryos therefore occur at the organizer of the mid-gastrula and we conclude that this is the stage and site of the primary function of *Zic2*.

This defect however, occurs after the organizer has finished producing the majority of the PCP cells. Accordingly, the cells that are under-produced appear to primarily be those that should contribute to the anterior notochord (Fig. 5K–O). Interestingly, *Zic* related genes have been implicated in the formation

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of the anterior notochord in ascidian embryos, suggesting that amongst the vertebrate Zic genes it may be *Zic2* which has retained this ancestral function (39,40). A link between the dearth of anterior notochord cells in $Zic2^{Ku/Ku}$ embryos and subsequent failure of PCP development is provided by the embryological experiments of Camus *et al*. (34). In these experiments, a region of the anterior mesendoderm was ablated and the consequence examined after a period of *ex utero* development. Removal of the PCP precursors was followed by reconstitution of the PCP, whereas removal of the cells fated to become anterior notochord lead to a failure of PCP development. These experiments imply that the maintenance and forebrain patterning activity of the PCP is dependent upon the presence of the anterior notochord. The kumba mouse mutant provides genetic evidence for this two-step mode of PCP development and serves as an excellent model for future investigations of this interaction between the anterior notochord and the PCP.

The work presented here provides an insight into the variable expressivity which is known to accompany human HPE. All of the embryos examined here which carried a mutation only in *Zic2* were congenic on the C3H/HeH background and therefore genetically identical. Despite this, the mutant embryos exhibited extreme phenotypic variability when examined at 12.5 dpc. This phenotypic variability increased with the age of embryos examined; embryos up to the neural plate stage of development tended to show little variation for a given marker assay, but by the 9.5 dpc stage the phenotypic variability was large (Fig. 3A–D). A possible explanation for this is that small differences at early stages are amplified by the stochastic nature of midline formation. One intriguing finding is the variability in the amount and location of notochord cells at headfold and later stages of development (compare the T expression in Figs. 3R and 4H). Axial mesoderm cells move along the axis once they emerge from the node and it appears that in $Zic2^{Ku/Ku}$ embryos their location at the headfold stage is not entirely indicative of their later location. Additionally, the relatively normal dorsal ventral patterning of $Zic2^{Ku/Ku}$ neurectoderm previously described (26) and shown here (Fig. 3G–J) is unexpected. It seems that relatively few notochord/PCP cells are required to establish initial neurectoderm pattern. Our experiments also modified the genetic background of the *Zic2* mutation (a congenic 129/SvEv $Zic2^{Ku/Ku}$ strain was used for the *Shh* genetic interaction studies), but we found this had little effect on the range or proportion of phenotypes seen. This may, however, be a coincidence rather than evidence that genetic variation does not greatly influence the described phenotypic variability. We have observed that $Zic2^{Ku/Ku}$ heterozygous animals exhibit *spina bifida* on the BALB/c background but not on the C3H/HeH or 129/SvEv background and it may be that the BALB/c background would also alter the penetrance and expressivity of HPE.

In summary, this work in combination with other recently published data (8) provides an explanation of *Zic2*-associated HPE. *Zic2* plays two distinct roles in forebrain development. At mid-gastrulation *Zic2* functions to maintain the organizer region and severe loss-of-function mutations lead to a transient failure in anterior mesendoderm production which in turn leads a failure to establish the ventral forebrain signalling centre and gross perturbations in forebrain D–V pattern. Apparently this requirement is well buffered such that mild loss-of-function is compatible with correct function of the ventral forebrain signalling centre. A second requirement for *Zic2* function occurs in the dorsal neurectoderm of the developing brain where it acts to promote formation of the dorsal signalling centre responsible for roof-plate and choroid plexus development. This requirement is more stringent, since hypomorphic mutations in both human and mouse reveal this function (6,9). Continued work will enable understanding of the genetic networks that *Zic2* utilizes at each of these two stages of forebrain development and so pave the way for attempts to ameliorate *ZIC2*-associated HPE.

MATERIALS AND METHODS

Mouse husbandry, strains and alleles

The kumba (*Ku*) allele of Zic2 ($Zic2^{Ku}$) (26) was maintained on two distinct backgrounds by either continuous backcross to C3H/HeH or 129/SvEv inbred mice. The *Shh* null allele ($Shh^{-/-}$) (17) and the smoothened null allele ($Smo^{-/-}$) (41) were maintained by continuous backcross to 129/SvEv mice. In all cases mice derived from backcross 10 and beyond were used for analysis. Analysis of the $Zic2^{KuKu}$ phenotype was performed using embryos derived from the congenic C3H/HeH colony, whereas the *Shh* interaction study was performed on the 129/SvEv background. Animals of the genotype $Shh^{+/+}$;$Zic2^{Ku/+}$ and $Shh^{+/-}$;$Zic2^{+/+}$ were intercrossed to produce $Shh^{+/-}$; $Zic2^{Ku/+}$ *trans*-heterozygote animals. *Trans*-heterozygous females and males were intercrossed to produce $Shh^{-/-}$;$Zic2^{Ku/Ku}$ double mutant embryos as well as all other compound genotypes. Mice were maintained in a light cycle of 12 h light:12 h dark, the midpoint of the dark cycle being 1 A.M. For the production of staged embryos, 1 P.M. on the day of appearance of the vaginal plug is designated 0.5 dpc. Mice were genotyped by PCR screening of genomic DNA extracted from ear biopsy tissue and embryos were genotyped using a fragment of extra embryonic tissue/ectoplacental cone or yolk sac (depending on stage). Genomic DNA was extracted as described previously (42). The $Zic2^{Ku}$ allele was identified using an allelic discrimination assay (26). The *Shh* null allele was identified using a novel three-primer multiplexed PCR with an annealing temperature of 62°C and 34 cycles. Primers used were as follows: PGKR, 5′-CCATTTGT CACGTCCTGCACGA-3′; ShhF, 5′-CACACACACATTTCT CTGTCCA-3′; ShhR, 5′-TAGCTCAGTGCTTGCAAGGTT A-3′. These primers generate amplicons of 249 and 260 bp from the wild-type and Shh null alleles, respectively. The *Smo* null allele was genotyped as described previously (41).

Whole-mount *in situ* hybridization

Procedures for embryo collection and WMISH have been described previously (26). Embryos were staged according to Downs and Davies, 1993 (43). This system uses the following abbreviations: PS, pre-streak stage, ES, early steak stage; LS, late streak stage; OB, no bud stage; EB, early bud stage; LB, late bud stage; EHF, early headfold stage; LHF, late headfold stage. For each probe and embryonic stage examined, between 4 and 15 mutant embryos were subjected to WMISH and

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compared with a similar group of stage-matched controls. Full details of the numbers of embryos examined are provided in Supplementary Material, Table S1. Probes for the following genes have been described previously: *Cer1* (33), *Chrd* (44), *Fgf8* (45), *Foxa2* (46), *Foxg1* (47), *Gli1* (48), *Gsc* (49), *Hesx1* (50), *Hhex* (51), *Lhx1* (52), *Otx2* (53), *Shh* (31) and *T* (54). *Oct4*, *Ptch1*, *Six3* and *Titf1* probes were generated from the following I.M.A.G.E Consortium [LLNL] cDNA Clones (55). *Oct4*: IMAGE:906785, Acc no. AA522407, (*BamH*I, T7 polymerase). *Ptch1*: IMAGE:3972649, Acc no. BF719201, (*EcoR*I, T3 polymerase). *Six3*: IMAGE:761326, Acc no. AA388136, (*Sal*I, SP6 polymerase). *Titf1*: IMAGE: 6416507, Acc no. BC057607, (*EcoR*I, T3 polymerase).

Histology

Embryos for haematoxylin and eosin or Ab staining were wax embedded. They were fixed in 4% PFA in PBS at 4°C before dehydration through a methanol series (25, 50, 75 and 100%×3 changes) followed by three changes of the Xylene substitute 100% Sub-X (Surgipath). They were transferred to base moulds and underwent five paramat wax (VWR) changes before orienting and embedding. Sections (7 μm) were cut and collected onto charged slides (BDH). Haematoxylin and eosin staining was performed according to standard protocols. Frozen sections (15 μm) were cut from embryos that had undergone WMISH using procedures described previously (26).

Immunohistochemistry and cell counting

Sections were de-waxed and re-hydrated using standard procedures and then boiled in 10 mM Sodium Citrate for 14 min, incubated in 1% H_2O_2 for 10 min at RT, blocked in 5% bovine serum albumin and then incubated overnight at 4°C with the following primary antibodies: Cleaved Caspase 3 (Asp175) anti-rabbit 1:200 (Cell Signaling Technology) and Monoclonal anti-phospho-histone H3 pSer (28) anti-rat 1:1000 (Sigma). Sections were then washed three times in PBS for a total time of 30 min and subsequently incubated for 1 h at room temperature in the following secondary antibodies: Alexa fluor 488 goat anti-rabbit IgG (H+L) 1:1000 (Molecular Probes) and Alexa Fluor goat anti-rat IgG (H+L) (Molecular Probes). Sections were washed in PBS overnight and mounted in Vectashield with DAPI (1.5 μg/ml) (Vector Labs). Sections were viewed in a compound microscope (Zeiss Axiophot 2) with epiflourescence using filters for wavelengths 488 (Green), 594 (Red) and 350 (Blue). The cells undergoing apoptosis (Caspase 3 positive) or cell proliferation (histone H3 positive) were counted in all sections through the embryo from each layer of the embryo (ecto-, meso- and endoderm). Digital images were taken at each wavelength and merged in Adobe Photoshop 7.0.

SUPPLEMENTARY MATERIAL

Supplementary Material is available at HMG Online.

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